



Course Title : Computer Networking Lab

Course Code : CSE342

Section : 1

Project

Project Title : Implementing DHCP and DNS Services.

Submitted By :

Name : Md. Raihan Shikder

ID : 2021200000089

Batch : 59

Department of CSE

Submitted To :

Name : Mr. Eamir Ahmed

Senior Lecturer, Department of
Computer Science & Engineering
Southeast University

Table of Contents

1. Introduction
2. Objectives
3. Network Topology
4. Configuration Steps
 - 4.1. Router Configuration
 - 4.2. DHCP & DNS server configuration
 - 4.3. Switch configuration
 - 4.4. PC configuration
5. Testing the configuration
6. Conclusion

Introduction

Cisco Packet Tracer is a network simulation tool developed by Cisco Systems that allows users to create, configure, and troubleshoot virtual networks. It is used for educational purposes. Students can learn about networking and demonstrate them.

Implementing Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) services with the help of Cisco Packet Tracer. The objective is to create a functional network that automates IP address allocation through DHCP and resolves domain names via a DNS server. By configuring these, it will make network management easy and more efficient.

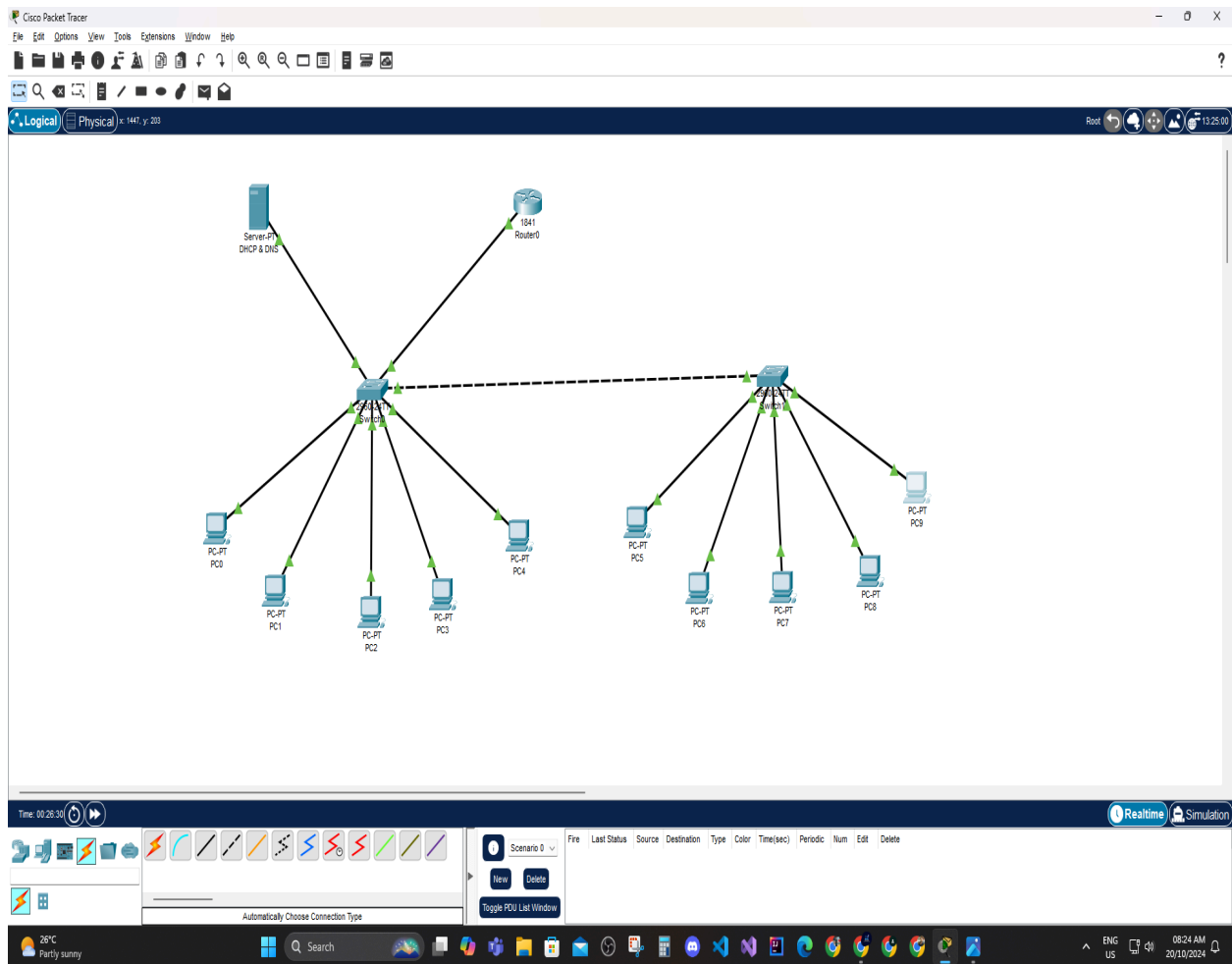
Objectives

1. To configure a DHCP server to automatically assign IP addresses to PCs.
2. To set up a DNS server to resolve domain names to IP addresses.

Network Topology

Devices and Tools:

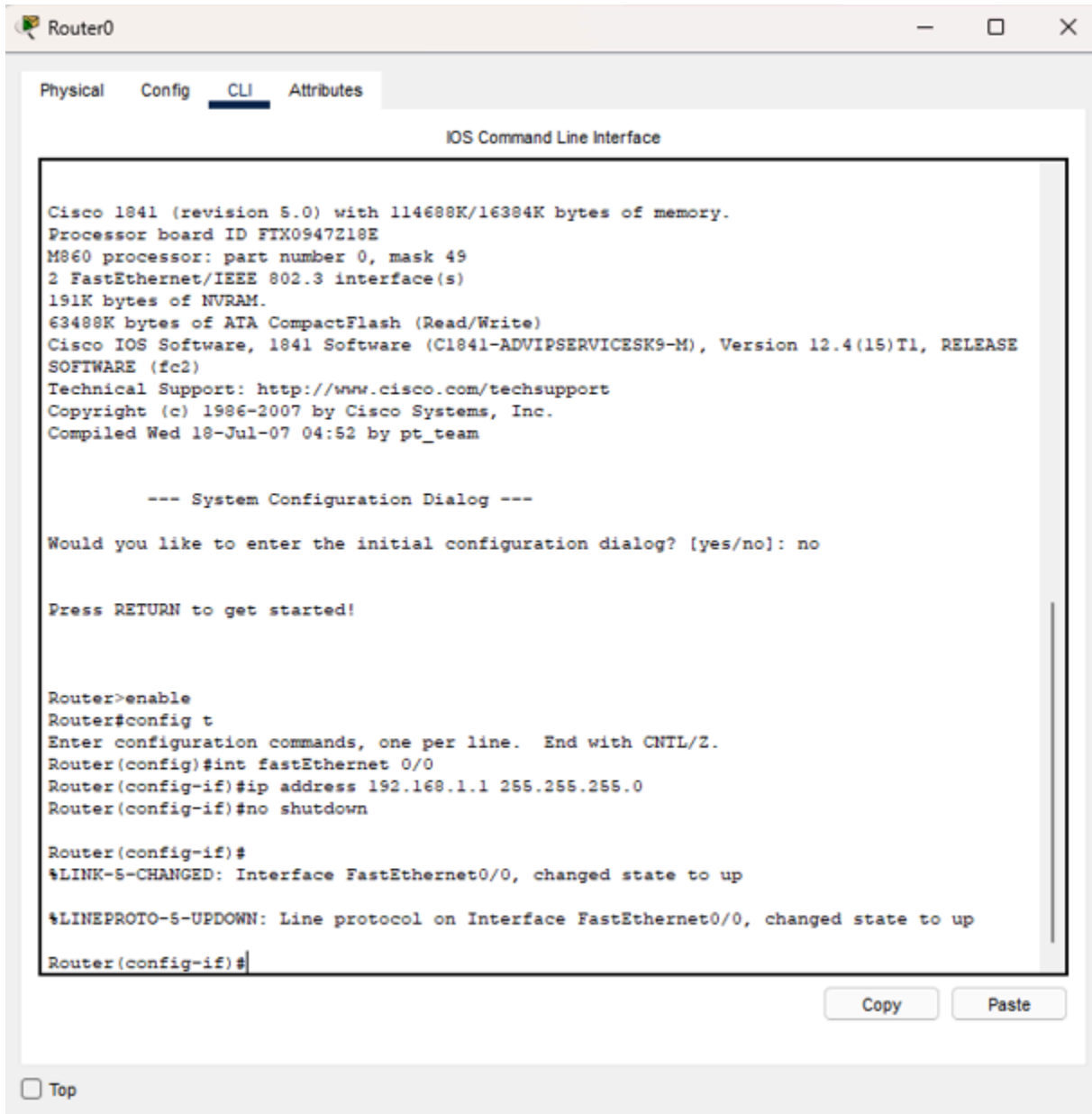
Router	1
DHCP & DNS Server	1
Switches	2
PCs	10



Configuration

Step 1: Open Cisco Packet Tracer.

Step 2: Then 1 Router taken and configured.



Ip address given to the router is: 192.168.1.1

Subnet Mask: 255.255.255.0

Step 3: 2 Switches have been taken. 1 Switch is connected with the router and both switches are connected to each other with cross-over cable.

Step 4: 1 server is taken. Set up the server for DHCP and DNS.

The screenshot shows a window titled "DHCP & DNS" with tabs for Physical, Config, Services, Desktop, Programming, and Attributes. The "Config" tab is active, and the "IP Configuration" section is expanded. It contains three sub-sections: IP Configuration, IPv6 Configuration, and 802.1X.

IP Configuration

- ☐ DHCP
- ☒ Static
- IPv4 Address: 192.168.1.2
- Subnet Mask: 255.255.255.0
- Default Gateway: 192.168.1.1
- DNS Server: 0.0.0.0

IPv6 Configuration

- ☐ Automatic
- ☒ Static
- IPv6 Address: [empty] / [empty]
- Link Local Address: FE80::20D:BDFE:FE96:A1BD
- Default Gateway: [empty]
- DNS Server: [empty]

802.1X

- ☐ Use 802.1X Security
- Authentication: MD5
- Username: [empty]
- Password: [empty]

☐ Top

Ip configuration of the server is given.

Server Ip Address: 192.168.1.2

Subnet Mask: 255.255.255.0

Default Gateway: 192.168.1.1

Step 5: Turning On the DHCP & DNS of the server. Then configured the server. Given Addresses are:

Pool Name: serverPool.

Default Gateway: 192.168.1.1

DNS server: 192.168.1.2

Start IP: 192.168.1.0

Subnet Mask: 255.255.255.0

Physical Config **Services** Desktop Programming Attributes

SERVICES

- HTTP
- DHCP**
- DHCPv6
- TFTP
- DNS
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DHCP

Interface: FastEthernet0 Service: ☒ On ☐ Off

Pool Name: serverPool

Default Gateway: 192.168.1.1

DNS Server: 192.168.1.2

Start IP Address: 192 168 1 0

Subnet Mask: 255 255 255 0

Maximum Number of Users: 256

TFTP Server: 0.0.0.0

WLC Address: 0.0.0.0

Add Save Remove

Pool Name	Default Gateway	DNS Server	Start IP Address	Subnet Mask	Max User	TFTP Server	WLC Address
serverPool	192.168....	192.168....	192.168....	255.255....	256	0.0.0.0	0.0.0.0

☐ Top

Saved.

Step 6: Then turning On the DNS option and given a Domain name and an address to it.

Domain Name: netlab.com

Address: 192.168.1.2

The screenshot shows the 'DHCP & DNS' configuration window. The 'Services' tab is selected, and the 'DNS' service is turned on. A resource record is added for the domain 'netlab.com' with the IP address '192.168.1.2'.

SERVICES

- HTTP
- DHCP
- DHCPv6
- TFTP
- DNS**
- SYSLOG
- AAA
- NTP
- EMAIL
- FTP
- IoT
- VM Management
- Radius EAP

DNS

DNS Service ☒ On ☐ Off

Resource Records

Name Type

Address

No.	Name	Type	Detail
0	netlab.com	A Record	192.168.1.2

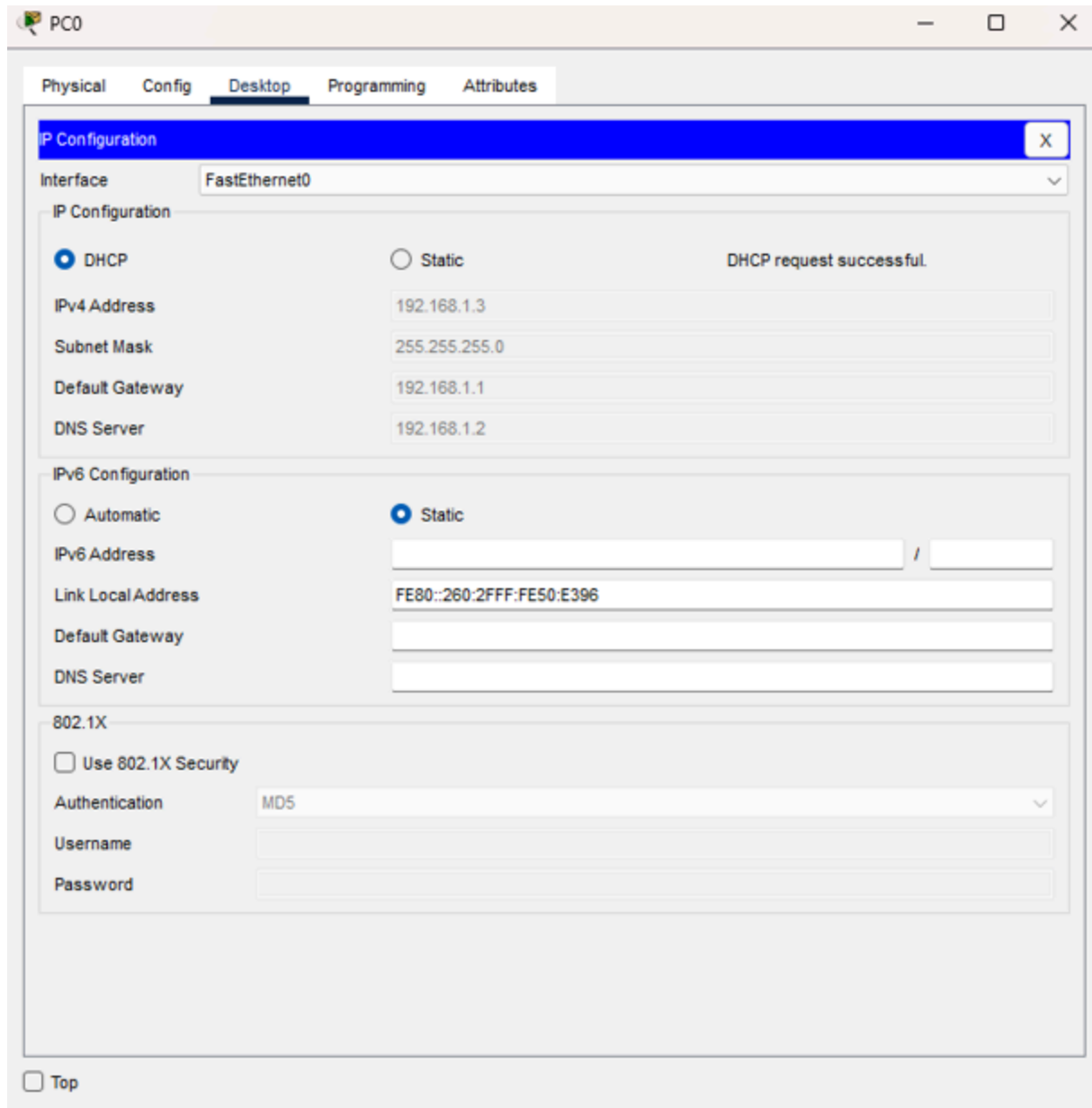
☐ Top

Save it.

Step 7: Then taken 10 PCs. 5 PC is connected to 1 switch and 5 PC is connected to another switch.
Now Configured the ip address of every PCs in the same way.

The screenshot shows the configuration window for PC0, specifically the 'Desktop' tab. The 'IP Configuration' section is highlighted in blue. The 'Interface' dropdown is set to 'FastEthernet0'. Under 'IP Configuration', the 'DHCP' radio button is selected, and the 'Static' radio button is unselected. The 'Requesting IP Address' checkbox is checked. Below this, there are input fields for 'IPv4 Address', 'Subnet Mask', 'Default Gateway', and 'DNS Server'. The 'IPv6 Configuration' section is also visible, with the 'Automatic' radio button unselected and the 'Static' radio button selected. The 'IPv6 Address' field is empty, and the 'Link Local Address' field is populated with 'FE80::260:2FFF:FE50:E396'. Below the IPv6 section, there is a '802.1X' section with a 'Use 802.1X Security' checkbox, an 'Authentication' dropdown set to 'MD5', and input fields for 'Username' and 'Password'. A 'Top' button is located at the bottom left of the window.

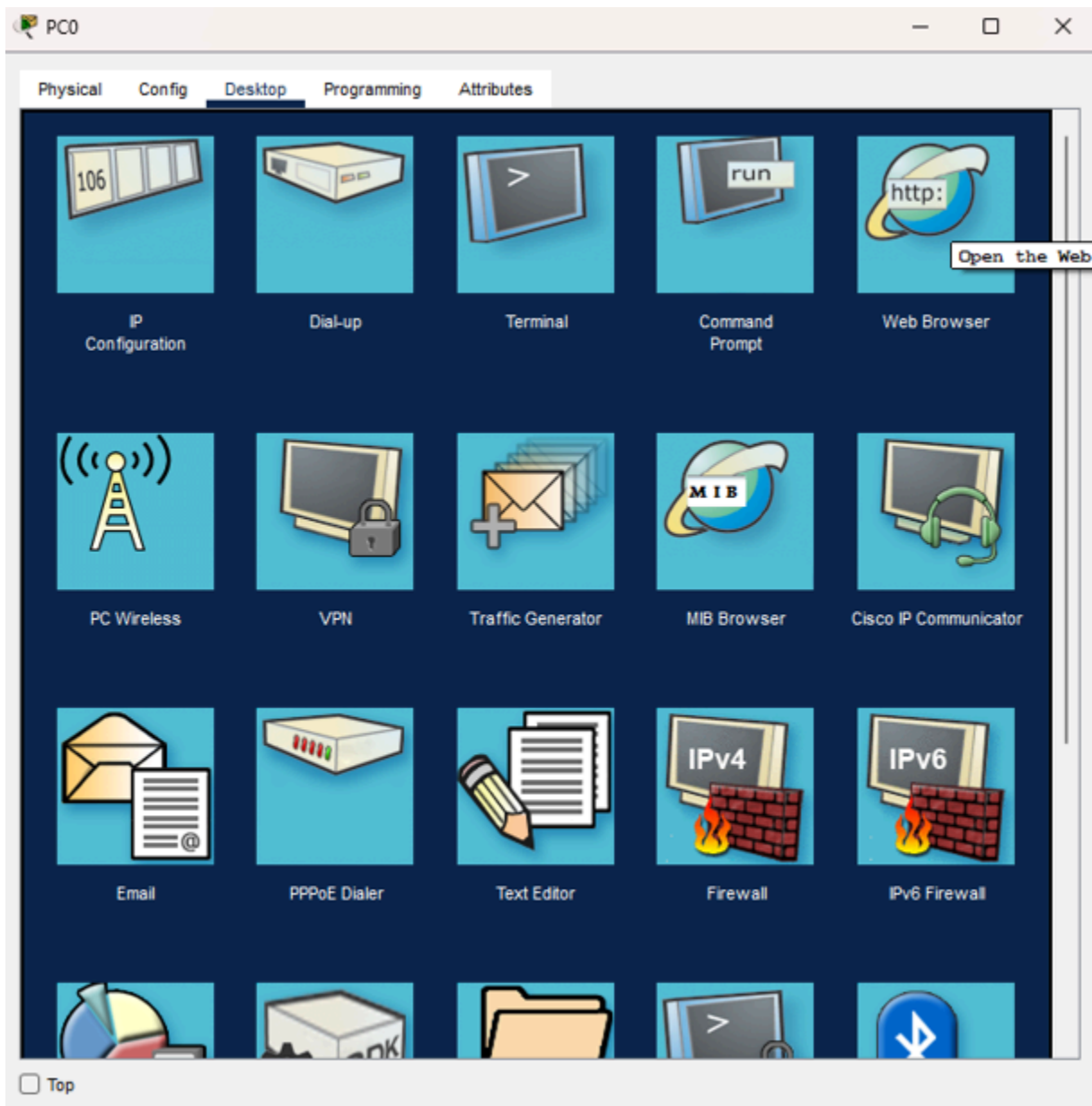
When the DHCP option is clicked it is requesting for IP addresses.



Automatically IP addresses are given to the PC.
Same way other PCs are given IP Addresses.

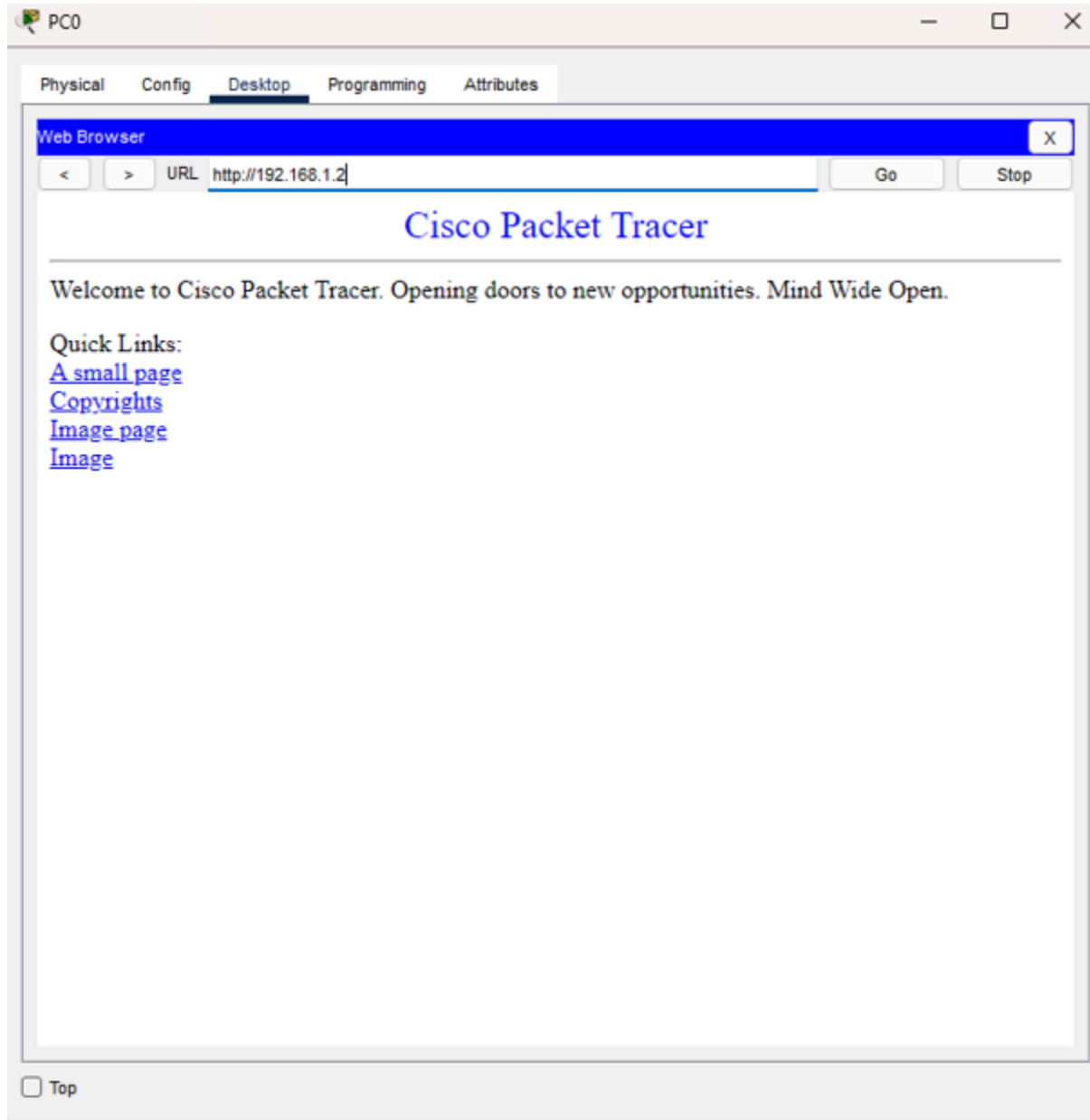
Testing Configuration

Step 1: Then checked if the server is responding or not.



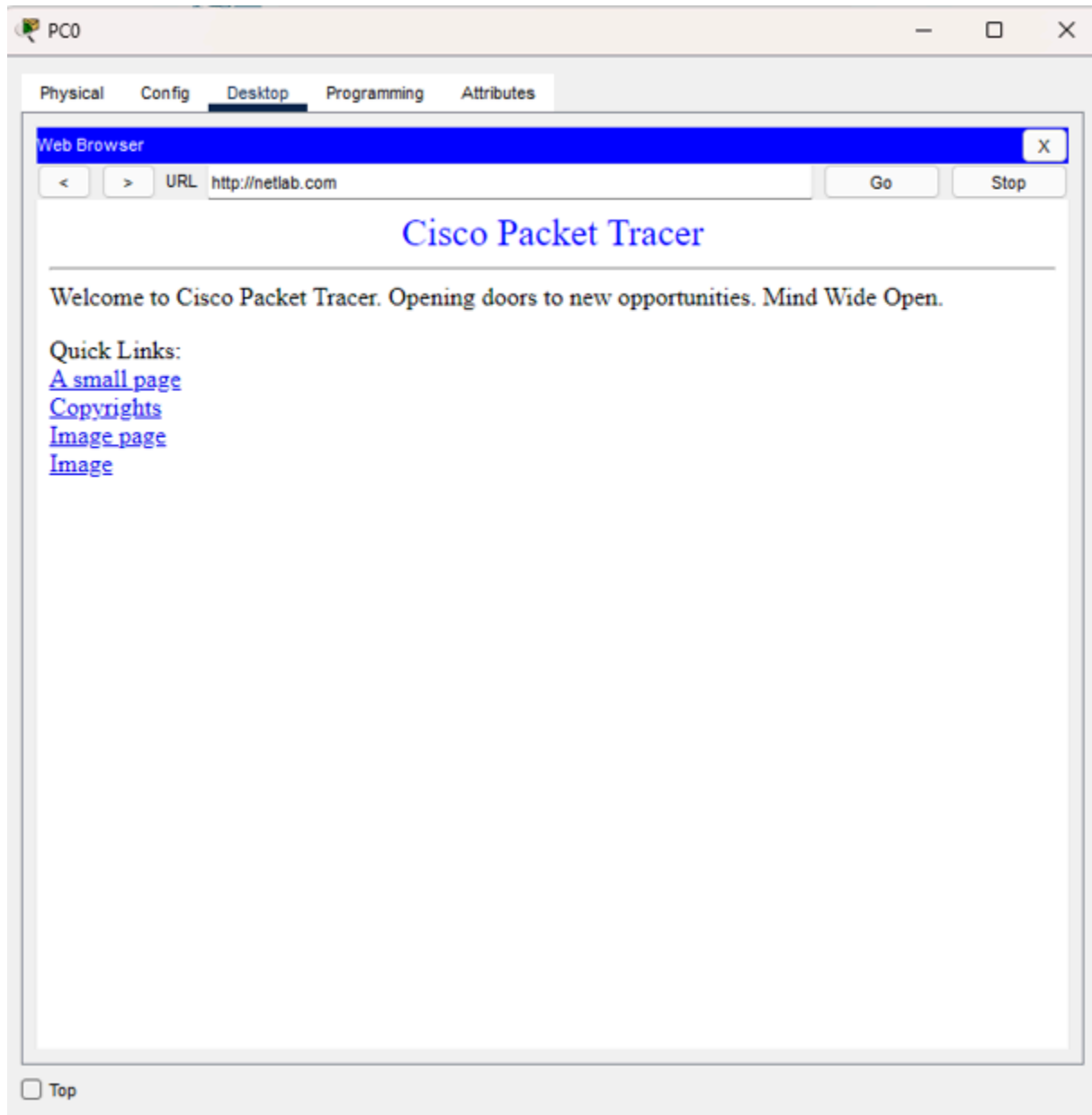
Goes to Web Browser.

Then typed the server address: 192.168.1.2



The server is responding with the IP address.

Step 2: Then check with the Domain Name that was given to the server.



The server is responding with the Domain Name.

The IP Addresses are:

Device	IP Address	Subnet Mask
Router	192.168.1.1	255.255.255.0
Server	192.168.1.2	255.255.255.0

PCs IP addresses:

Devices	IP Address	Subnet Mask	Default Gateway	DNS
PC0	192.168.1.3	255.255.255.0	192.168.1.1	192.168.1.2
PC1	192.168.1.4	255.255.255.0	192.168.1.1	192.168.1.2
PC2	192.168.1.5	255.255.255.0	192.168.1.1	192.168.1.2
PC3	192.168.1.6	255.255.255.0	192.168.1.1	192.168.1.2
PC4	192.168.1.7	255.255.255.0	192.168.1.1	192.168.1.2
PC5	192.168.1.8	255.255.255.0	192.168.1.1	192.168.1.2
PC6	192.168.1.9	255.255.255.0	192.168.1.1	192.168.1.2
PC7	192.168.1.10	255.255.255.0	192.168.1.1	192.168.1.2
PC8	192.168.1.11	255.255.255.0	192.168.1.1	192.168.1.2
PC9	192.168.1.12	255.255.255.0	192.168.1.1	192.168.1.2

Conclusion

This project successfully implemented DHCP and DNS services in Cisco Packet Tracer, demonstrating their vital roles in network management. By automating IP address allocation and simplifying domain name resolution, we improved network efficiency and usability.